



Artificial intelligence (AI) and machine learning (ML) in procurement and purchasing decision-support (DS): a taxonomic literature review and research opportunities

Dursun Balkan¹ · Goknur Arzu Akyuz²

Accepted: 22 July 2025 / Published online: 20 August 2025
© The Author(s) 2025

Abstract

Artificial intelligence (AI), machine learning (ML) and decision-support (DS) are gaining increasing interest with widening adoption. This article investigates the enabler role of AI and ML for providing decision-support in procurement&purchasing domain. The study follows a systematic review approach via taxonomic analysis. Comprehensive analysis and discussions are provided for: (a) the relevance and applicability of AI and ML in procurement&purchasing decision-support; (b) functionalities/processes for which they are utilized; (c) related methodologies; and (d) implementation benefits as well as challenges. Findings reveal that procurement&purchasing area holds significant potential in terms of AI-ML applications for decision-support almost every related sub-process. This study is original by offering a process-oriented approach to the research domain; providing unique clustering and classification; and presenting detailed analyses via unique taxonomy tables with respect to approach, topic, focus, context and methodologies of the literature items reviewed. The study offers further research opportunities and has significant potential to provide managerial insights by the identified sectoral applications, benefits and challenges.

Keywords Artificial intelligence · Machine learning · Decision support systems · Procurement · Purchasing

✉ Goknur Arzu Akyuz
arzu.akyuz@gmail.com

Dursun Balkan
dursunbalkan@gmail.com

¹ Department of Management Information Systems, Gazi University, Ankara, Turkey

² Department of Management Information Systems, University of Turkish Aeronautical Association, Ankara, Turkey

1 Introduction

With the operational connectivity problems overcome via a variety of Internet of Things (IoT) technologies, information systems are assuming an increasingly strategic role in supply chain management (SCM). Providing tactical and strategic-level decision-support (DS) integrated with the backbone enterprise resources planning (ERP) systems becomes the major concern in all industries and contexts. Relying on diverse sets of data coming from internal as well as external systems, DS systems utilize a wide spectrum of modeling and analysis tools, user interfaces and collaboration tools to cater for decision-making needs of people. In this context, current developments in DS and SCM aim at integrating supplier relationship management (SRM), ERP and customer relationship management (CRM) solutions (Kontogounis et al. 2006), to encompass and assist inter and intra-enterprise processes.

In today's IT-intensive business ecosystems, artificial intelligence (AI) and machine learning (ML) are two key recent advances which are proven to redefine the business processes, and they are well-supported to be a driving force for process improvement (Allal-Cherif et al., 2021; Guida et al. 2023).

Role and importance of AI in SCM is clearly highlighted in the literature for decision-making, facilitating cross-partner collaboration, supporting proactive strategy development and acting as a strategic asset for competitiveness (Hendriksen 2023; Richey et al. 2023). AI can automate standard activities, replace human intelligence with some unique skills remaining preferable for some activities, enhance human intelligence; act as a collaborative partner with autonomous capabilities and support proactive process management (Burger et al. 2023; Hendriksen 2023). In this regard, development of hybrid intelligence is in question. ML is generally accepted in literature as a significant subset of AI, focusing on the algorithms learning from data and encompassing supervised, unsupervised and reinforcement learning algorithms. Although AI and ML can be used interchangeably in some cases, they are conceptually different, and AI represents a broader category. In this study, these two concepts are treated simultaneously in relation to purchasing for capturing and covering the body of knowledge within the specific subset of ML as well as the broader scope of AI.

Procurement process, which is the key to SCM, is gaining increasingly strategic character, and obtaining procurement-related decision-support becomes vital for tactical and strategic levels. Transformation witnessed in the procurement area is increasingly data-driven, characterized by the highlighted importance of predictive capabilities and analytics, with a shifting focus towards strategic decision-making and increased need for visibility, scalability and adaptability. Hence, procurement is increasingly turning to AI and ML among various other business processes (Bodendorf et al. 2022a, b; Papachristou et al. 2021; van Hoek 2024). Bringing in automation and standardization, handling big data, improving predictive capabilities, enabling spend compliance, enabling data-driven decisions facilitating better adaption capabilities to new trends, and increasing smartness are the basic promises of AI and ML to procurement domain. Despite being widely adopted (Burger et al. 2023; Bodendorf et al. 2022b; Kuo et al. 2021), the intersection of AI and procurement is still immature, with low levels of adoption and significant future potential (Guida et al. 2023; van Hoek 2024). Thus, AI and ML are among the most significant, disruptive and contemporary innovations holding a transformative potential to revolutionize the procurement process, of which potential benefits are not fully realized yet.

Motivated by this transformative potential, this research aims to provide a systematic literature review at the intersection of AI, ML, procurement and decision-support concepts. The study answers the following research questions in relation to **procurement&purchasing decision-support**:

R1 *How do AI and ML act as the main enablers?*

R2 *What are the main sectoral applications of AI and ML?*

R3 *What are the functions/processes for which AI and ML are most frequently used?*

R4 *What are the benefits of using AI and ML?*

R5 *What are the challenges of using AI and ML?*

Contributions of this study are as follows:

- a) Simultaneous treatment of AI, ML, DS and procurement&purchasing concepts, which are under-investigated hot topics,
- b) Providing a comprehensive and systematic review approach at the intersection of these multiple themes by organizing and synthesizing existing knowledge in this advancing field,
- c) Offering a detailed taxonomic analysis of the identified literature items under the dimensions of approaches and methodologies used, related SCM processes/functionalities, sectoral breakdown of the applications and benefits,
- d) Following a process-oriented approach to map the utilization of ML and AI to different functionalities/processes,
- e) Providing managerial insights in terms of a variety of sectoral applications, benefits and challenges.

The study has a multi-disciplinary character, contributing simultaneously to AI, ML and procurement disciplines. It is appealing for researchers and specialists from procurement, supply chain, data science, statistics AI and ML domains. Thus, it is valuable for academicians as well as practitioners.

Sections for the remainder of the study are as follows: Sect. 2 describes the systematic literature review methodology. Section 3 presents the summary analytics for the review. Section 4 offers clustering approach. Section 5 offers analysis with respect to main approach and main topic. Section 6 provides the clustering and analysis with respect to context, main focus and methodology. Section 7 contains the analysis by functionalities/processes. Section 8 analyses the related benefits and challenges. Section 9 provides discussion and further research avenues, and Sect. 10 concludes.

2 Literature review methodology

Our literature review is conducted on Web of Science (WoS) as the main database. It is preferred to focus on the Web of Science database for providing clear and focused coverage in our systematic review methodology, avoiding duplications across the databases, and ensuring the quality of the resources utilized by excluding various web sources which are not of equal quality.

The flowchart describing the searching and filtering process is in Fig. 1, and steps are described below:

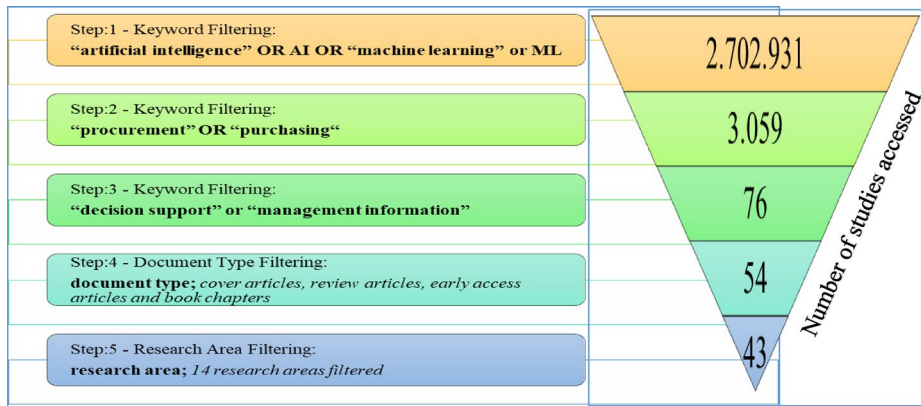


Fig. 1 Systematic review methodology

Step 1. Initial WoS search is performed with the keywords:

"artificial intelligence" OR AI OR "machine learning" or ML.

Step 2. Within the search results, further filtering is applied with the keywords:

"procurement" OR "purchasing".

Step 3. Further filtering is made with the keywords:

"decision support" or "management information".

Step 4. Document type filter is applied to cover articles, review articles, early access articles and book chapters. Other document types are excluded.

Step 5. Filtering is done according to the most relevant 14 research areas (given in Table 2).

This systematic is designed to capture the studies focusing on specific ML algorithms as well as broader AI scope. Conference papers are excluded since majority of them were accessible only as abstract, not full text. No date filter is applied during searching and filtering to provide comprehensive coverage.

After filtering down to 43 studies, one article is found to be retracted, and excluded from the analysis. Therefore, analysis is conducted over 42 articles. Identified articles are analyzed systematically via taxonomic analysis in the subsequent sections.

3 Literature review summary analytics

Based on the above-described literature review methodology, this section offers summary analytics for the identified studies. Bibliometric analysis results are given in Figs. 2, 3 and 4; Tables 1 and 2.

Distribution of the number of articles by year is in Fig. 2. This graph clearly indicates the suddenly increasing interest in the intersection of the topics studied, especially starting year 2021. It is striking to observe that total number of studies in the time interval of 15 years during 2002–2017 is 13, increasing to 30 within last 5-years' time interval of 2019–2023. This spotlights the dramatically increasing importance of AI and ML for decision-support after 2020's. This can be attributable to various factors such as growing AI capabilities, increasing strategic importance of procurement and increased need for real-time decision-making and the transformative benefits of the technology. It is important to note that the beginning of 2020's corresponds to radical advancements in AI and ML, especially in the areas of reinforcement learning, large language models, multimodal AI, and AI applications in different industries.

Distribution of resources by countries (based on authors' origins) is in Fig. 3. It appears that USA and France have the highest frequencies. This variation in terms of the author's origins can be due to a number of factors such as academic ecosystem, technological and computational resources and funding availabilities. Since the research topic is highly interdisciplinary, these differences can also be attributable to the collaborative research culture and capabilities of the countries.

Distribution of resources with respect to the citation indices is offered in Fig. 4. It is clear that majority of the studies are from SCI-Expanded and SSCI-indexed resources.

Shows the distribution of the studies with respect to publishers. According to the table, Elsevier has the highest count of relevant studies in our search, followed by Springer and MDPI.

Distribution of the studies with respect to 14 selected research areas is given in table 2. Related studies are published mostly in the areas related to computer science, engineering, operations research management science, and business economics. Please note that in this WoS-based distribution, one Article May be reported by WoS in more than one research area

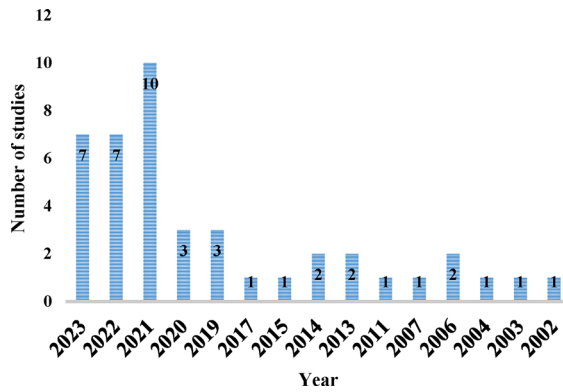


Fig. 2 Number of studies by year

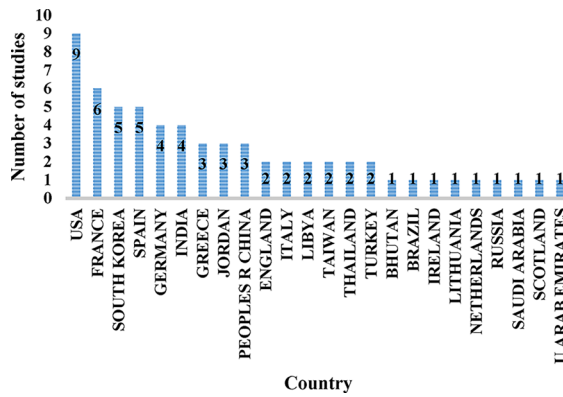


Fig. 3 Distribution of studies by countries

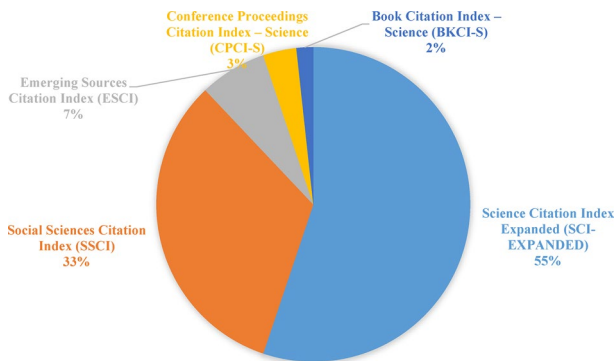


Fig. 4 Distribution of studies with respect to indices

Table 1 Distribution of studies with respect to publishers

Publisher	Count
Elsevier	15
Springer Nature	8
Mdpi	7

Table 1 Distribution of studies with respect to publishers

Publisher	Count
Emerald Group Publishing	2
Taylor & Francis	2
Asce-Amer Soc Civil Engineers	1
Hindawi Publishing Group	1
IEEE	1
Igi Global	1
Inst Engineering Technology-Iet	1
Omniascience	1
Oxford Univ Press	1
Sage	1
Soc Inform Manage-Mis Res Cent	1

Table 2 Distribution of studies with respect to research areas

Research Areas	Count
Computer Science	21
Engineering	16
Operations Research Management Science	12
Business Economics	8
Environmental Sciences Ecology	5
Science Technology Other Topics	5
Automation Control Systems	3
Construction Building Technology	2
Information Science Library Science	2
Health Care Sciences Services	1
Mathematical Methods In Social Sciences	1
Medical Informatics	1
Social Sciences Other Topics	1
Telecommunications	1

4 Clustering approach

In systematic reviews, identification of the main clusters under which taxonomy tables and the main sections of discussion are organized is critical. This section describes the clustering rationale, and the taxonomic approach utilized in the study.

After a critical analysis of the articles, the following three clusters are determined by considering topical similarity, content coverage and research characteristics:

- Cluster-1: Papers with review characteristics, classified as the “overview of the topic”: This cluster contains the identified research items that utilize systematic literature review as the research methodology.
- Cluster-2: Papers about SCM processes, classified under “inbound, outbound and overall SCM”: Papers directly related to SCM processes are grouped under this cluster. The cluster is further organized according to the macro supply chain processes treated in the literature item in terms of inbound, outbound and the overall supply chain processes.
- Cluster-3: Papers directly focusing on sectoral applications: This cluster groups the

literature items which directly contain specific sectoral applications. Further subclassification is made under: “manufacturing applications”, “construction applications” and “other applications (finance, medical, and air forces)”. This classification is preferred to put forward a sectoral perspective by providing a sectoral breakdown of the applications at the intersection of key research themes.

After identifying these clusters, taxonomic analysis is performed via comprehensive taxonomy tables with respect to:

- main approach and main topic.
- context, main focus and methodology.
- functionality and processes.
- benefits.

Each taxonomy table is manually created and organized according to the above-mentioned clusters and subgroups which are described above.

5 Analysis according to main approach and main topic

Distribution of the articles according to main approach and main topic is given in Table 3. The table contains the cluster, classification, author/year, main approach and main topic for each study.

Based on this table, frequency distribution with respect to the main approaches used is given in Fig. 5. This figure reveals that papers using statistical approaches as well as the ones supported by case studies are dominant. Various mathematical modeling and statistical approaches are utilized in combination with the decision support system (DSS) concept along with a wide spectrum of AI and ML algorithms. It is also observed that comprehensive review studies are relatively few when compared to the other type of approaches. Hence, it can be reckoned that the interdisciplinary field of study still needs further systematic review studies for providing insights to researchers and practitioners.

Figure 6 shows the distribution of the studies with respect to the main topic. Since more than one main topic can be treated simultaneously in one paper, the graph includes all possible intersection combinations of 4 main topics of AI, ML, Procurement and DS. ML-DS and AI-DS intersections appear as the most frequently appearing combinations.

Table 3 Distribution according to main approach and main topic

Cluster	Classification	Study	Main approach				Main topic			
			Literature review/ Exploratory	Math- ematical Modeling	Statistical analysis	Multi-criteria	Case study	AI	ML	Procurement Deci- sion- support
Cluster 1 - Review	Overview	Allal-Cherif et al., 2021	✓				✓	✓		
		Burger et al. 2023	✓				✓	✓		✓
		Guida et al. 2023	✓					✓		✓
		Bodendorf et al. 2022b	✓						✓	✓
Cluster 2 - SCM Processes	Inbound	Reyes-Moro et al. 2003				✓	✓	✓		✓
		Zhou and Chen 2023		✓		✓	✓	✓		
		Canelas et al. 2013		✓				✓		✓
		Guillaume et al. 2014		✓	✓			✓		✓
		Canelas et al. 2015		✓				✓		✓
		Rodrigues-Aquilar et al., 2004		✓				✓		✓
		Siciliani et al. 2023						✓		✓
		Kuo et al. 2021			✓		✓		✓	✓
		Nissen et al., 2006			✓			✓	✓	✓
		Singh and Tucker 2017					✓		✓	✓
Cluster 3 - Sectoral Applications	Outbound	Kumar et al. 2022			✓			✓	✓	
		Biswas et al. 2022						✓	✓	
		Xu et al. 2011		✓			✓	✓	✓	✓
		Bodendorf et al. 2022c		✓				✓		
		Kontogounis et al. 2006				✓			✓	✓
		Kazancıoğlu et al., 2023				✓		✓		
Cluster 3 - Sectoral Applications	Manufacturing	Bodendorf et al. 2022a			✓			✓	✓	
		Buyvol et al. 2023					✓		✓	✓
		Zhang et al. 2021					✓		✓	✓

Table 3 (continued)

Cluster	Classification	Study	Main approach				Main topic			
			Literature review/ Explanatory	Mathematical Modeling	Statistical analysis	Multi-criteria	Case study	AI	ML	Procurement Decision- support
Construction		Papachristou et al. 2021					✓	✓	✓	
		Anglou et al. 2021			✓				✓	✓
		van Steenberghe and Mes 2020			✓		✓		✓	
		Shehadeh et al. 2021			✓		✓		✓	✓
		Alshboul et al. 2022a			✓				✓	✓
		Kaklauskas et al. 2007				✓	✓	✓		✓
		Kim et al. 2014					✓	✓		✓
		Kim et al. 2019	✓			✓	✓	✓		✓
		Yi et al. 2019			✓		✓	✓		✓
		Choi et al., 2022			✓		✓	✓	✓	
		Choi et al. 2021			✓		✓	✓		
		Alshboul et al. 2022b			✓				✓	
		Ahmed and El-Sayegh 2021	✓					✓		
Services (Finance, Medical, Air Force)		Leigh et al. 2002					✓	✓		✓
		Tuarob et al. 2021			✓				✓	✓
		Gomes et al. 2019				✓		✓	✓	✓
		Faric et al. 2023						✓		✓
		Owasirikul et al. 2013			✓		✓		✓	✓
		Caballero et al. 2023			✓				✓	✓

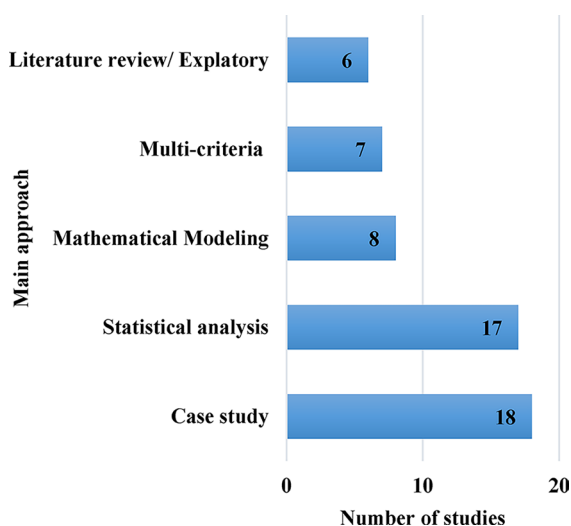


Fig. 5 Distribution of the studies with respect to the main approach used

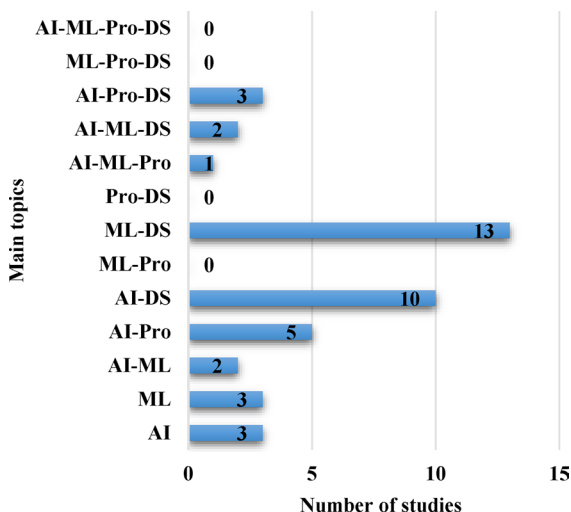


Fig. 6 Distribution of the studies with respect to the main topic

The following three subsections discuss the papers according to clusters identified.

5.1 Papers with review characteristics: overview of the topic

Procurement is one of the most fundamental processes in supply chain management. Starting with the needs identification, it covers all the stages of procure-to-pay cycle. A generic description of the procurement process is given in Fig. 7.

Literature items classified under the first cluster support the enabler role of AI and ML in almost all of these sub steps of the procurement cycle mentioned in Fig. 7. Automation and optimization of purchasing processes is one of the main reasons for using AI and ML (Allal-Cherif et al., 2021). Procure-to-pay cycle automation is crucial to enable real-time monitor-

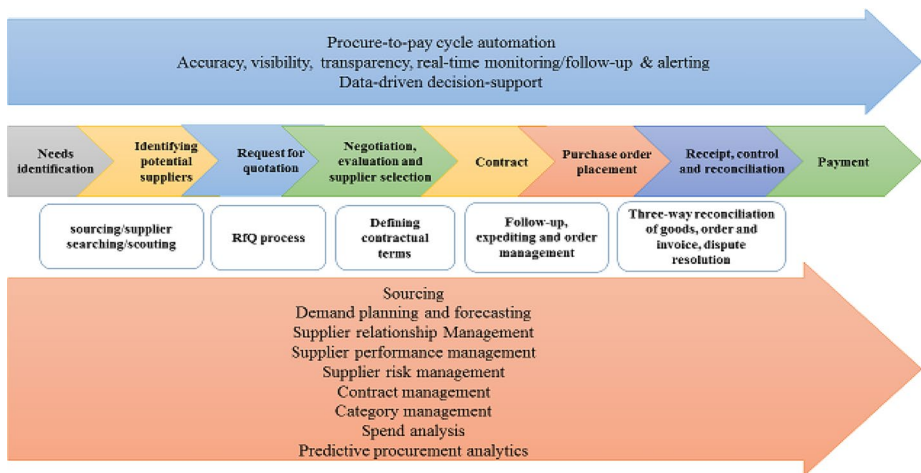


Fig. 7 A generic procurement process (based on Jonsson 2008; Guida et al. 2023)

ing, minimizing delays and mistakes, and obtaining accuracy in the overall cycle. Improved visibility and transparency, as well as monitoring and control over the processes appear as the most frequently cited benefits (Allal-Sherif et al., 2021; Burger et al. 2023; Guida et al. 2023; Bodendorf et al. 2022b; Reyes-Moro et al., 2003). Efficient sourcing decisions as well as advanced spend expenditure analysis (by segment, purchaser, program, client etc.) with advanced analytics are also facilitated by AI and ML applications.

It is observed that procurement process significantly benefits from advanced DS provided by AI and ML usage. Supplier selection via advanced matching systems facilitates automated evaluation of tenders, and support finding the best matching between parties (Canelas et al. 2013; Rodriques-Aquilar et al., 2004; Siciliani et al. 2023). Predictive buying and decision aids are assisted by big data analysis capabilities offered by AI and ML.

Supplier relationship management (SRM) processes are also supported by enabling controlling and follow-up with multi-criteria, not being limited to basic KPI sets. Rigorous real-time monitoring based on more complex and wide range of KPI's are possible. Burger et al. (2023) highlight demand prediction and planning, supplier selection, negotiation support, identifying and handling disruptions in order management, providing real-time incident alerts, as well as obtaining and managing procurement analytics and cost management as the main issues for which AI has significant potential. Study in Guida et al. (2023) also provides a comprehensive mapping of the application functionalities to the entire procurement cycle, and highlights chatbots, recommendation systems, and robotic process automation as the most commonly used AI applications.

Detailed discussion of challenges from procurement perspective is provided in Guida et al. 2023. A comprehensive review is also offered in Bodendorf et al. 2022b; discussing ML applications, challenges and obstacles supported by expert interviews, focusing on cost analysis and transparency.

Consequently, the review papers put forward the variety of functions/processes related to AI/ML intersection, benefits as well as implementation challenges involved. Utilization of diverse set of algorithms are also disclosed. These main issues will be elaborated in detail in the taxonomic analyses given in subsequent sections.

5.2 Papers related with SCM processes

During our analysis, nine papers classified as being related to the inbound side reveal that various subprocesses of end-to-end procurement cycle indicated in Fig. 7 (such as sourcing, supplier selection and supplier contracting subprocesses) frequently utilize AI and ML applications. E-application development for e-sourcing (Reyes-Moro et al. 2003), supplier selection in green context (Zhou and Chen 2023), DSS development for supplier selection process compliance to Sarbane-Oxley Act (Canelas et al. 2013), and supplier risk management with ill-known criteria (Guillaume et al. 2014) are the main studies that appeared in this context. Decision-support and web application development for contracting process (Canelas et al. 2015); industrial and public procurement (Rodrigues-Aquilar et al., 2004, Siciliani et al. 2023); smart support system development for obtaining predictive big data analytics for procurement (Kuo et al. 2021), and multi-agent based decision-support system development for procurement (Nissen et al., 2006) are the main studies supporting the applicability of AI and ML in relation to the inbound side. This group of papers clearly reveal that web-based and smart system development efforts are on-going for different aspects in various buying contexts.

Although the main focus is procurement and purchasing in this research, our review identified four studies (classified as “outbound”) supporting that AI and ML have significant applications in relation to the customer side for analyzing on-line customer product reviews and consumer opinions in e-commerce context (Singh and Tucker 2017; Kumar et al. 2022; Biswas et al. 2022; Xu et al. 2011). Undoubtedly, changing the keywords to the outbound side will definitely reveal details of the applications in this regard.

Three papers classified under “overall SCM” focus on applications in more than one SCM process. Bodendorf et al. 2022c deals with collaborative cost management. Kontogounis et al. (2006) develops an agent-based decision-support system for managing multiple SCM processes. Study in Kazancoglu et al. (2023) highlights the importance of overall SC sustainability and resilience.

Consequently, papers in cluster 2 clearly show a wide range of applications for the inbound side, outbound side as well as the overall supply chain.

5.3 Papers related with sectoral applications

Papers classified under sectoral applications put forward the applicability of AI and ML in different sectoral settings.

Manufacturing applications identified by six studies cover a wide spectrum of sectors, including automobile, clothing, shipping industry, and electronic toy car production. Cost estimation (Bodendorf et al. 2022a), diagnostic defect analysis (Buyvol et al. 2023), sales data analysis in big data context (Zhang et al. 2021), grouping and combining similar products (Papachristou et al. 2021), forecasting spare parts needs (Anglou et al. 2021), and forecasting new product demand for product prelaunch (van Steenberg and Mes, 2020) appear as the main applications. This group of articles clearly highlights the improved predictive power obtained by different hybrid applications of AI and ML.

Our analysis uncovered a number of construction-related applications as another significant subcluster. Ten articles identified in relation to construction industry, heavy equipment and engineering procurement and construction (EPC) indicate that this sector is a critical

one for AI and ML applications. In this context, heavy construction equipment residual value prediction (Shehadeh et al. 2021; Alshboul et al., 2022a); on-line system for evaluating construction alternatives (Kaklauskas et al. 2007); DSS development for material selection (Kim et al. 2014); schedule and cost performance (Kim et al. 2019; Yi et al. 2019), big data-driven multi-process management (Choi et al. 2021); constructor risk analysis (Choi et al., 2022), and green building cost prediction (Alshboul et al., 2022b) are the identified areas of focus in relation to the construction sector.

Service applications are also identified related to finance (2 studies in relation to stock market prediction by Leigh et al. 2002 and Tuarob et al., 2021); medical (3 studies by Gomes et al. 2019; Faric et al. 2023; Owasirikul et al. 2013) and air force training (Caballero et al., 2023).

Consequently, sectoral applications identified show the diversity of the sectors in which AI-ML-DS intersection is used in relation to procurement&purchasing. Dominance of the construction-sector applications is evident in this table, and this can easily be attributable to the huge number of procurement items with enormous monetary values, which can justify the development of specific DSS systems based on AI and ML. It is important to highlight that only very few service-related applications are identified during our search (6 studies, 3 of which are in medical sector). Definitely, search with different keywords can elaborate on these applications.

6 Analysis by context, main focus and methodology

With the same clustering and classification logic explained in Sect. 4, Table 4 provides the distribution of the studies according to context, main focus and details of the methodology used.

“Context” column in this table clearly shows the diversity in relation to their study contexts. This column ranges from various subprocesses of procurement to stock market prediction and construction project management.

“Main focus” column of the table indicates that DSS systems utilizing AI and ML aid various sub-processes of procurement (tendering, analysis of offers, negotiation, supplier selection etc.), support project management activities, and provide fact-based, big data analytics to all the sub steps of the procurement cycle. The use of DSS-AI-ML combination for prediction/ estimation/forecasting purposes in relation to demand, cost, price, spare part needs, equipment residual lifetime as well as customer behaviors also appears frequent. Detection-related and diagnostic uses are also evident related to defects and fraudulent records. It is shown that such.

“Methodology used” column shows the hybrid use of a variety of methods and algorithms. Based on the main approaches classified, we witness a variety of algorithmic methods. Various multi-criteria decision-making (MCDM) methods such as: TOPSIS and VIKOR (Zhou and Chen 2023); MAUT (Reyes-Moro et al. 2003); COPRAS (Kaklauskas et al. 2007), MACHBETH (Gomes et al. 2019); and their combinations with fuzzy logic, such as Fuzzy DEMATEL (Kazancoglu et al. 2023); Fuzzy inference system+AHP (Kim et al. 2019) are frequently used. Inherently, decision-making in procurement&purchasing context involves the evaluation of alternatives against a set of multiple criteria, necessitating the use of various multi-criteria approaches. However, utilization of traditional MCDM approaches gen-

erally relies on expert judgment, static weight assignments, and deterministic data inputs, restricting their applicability in dynamic, data-intensive procurement environments and limiting the refresh frequencies of the results. In this connection, literature reveals that AI and ML provide the following complementary capabilities: (a) Automated data processing and feature extraction to process vast amounts of structured and unstructured procurement data, including supplier performance metrics, procurement records, financial indicators, and external market information, (b) dynamic weight determination, facilitating dynamic and context-aware weighting of decision criteria for adapting to changing organizational priorities and external conditions, (c) predictive modeling and risk assessment for forecasting the future supplier performance or procurement risks based on past data, (d) clustering and classification for pre-screening alternatives to reduce the decision space; and (e) within hybrid AI-MCDM frameworks, MCDM results generally feeding the AI/ML approaches to obtain an initial criteria hierarchy or evaluations which can be further refined and enhanced with AI and ML algorithms. Thus, it appears that AI and ML enhance MCDM by enabling dynamic, data-driven, and predictive decision-making capabilities.

We also witness the co-occurrence of various statistical approaches, both independently and jointly in the literature. Different statistical approaches with various forms of regression are supported, such as Linear regression, Principal Component Regression and Bayesian Ridge regression (Guillaume et al. 2014; Tuarob et al. 2021; Caballero et al. 2023). Multivariate (MANCOVA) and univariate (ANCOVA) variance analysis (Nissen et al., 2006) are also utilized. Clustering algorithms such as k-means clustering (van Steenberg and Mess., 2020; Anglou et al. 2021); artificial neural networks (ANN) (Bodendorf et al. 2022a); deep neural networks (Alshboul et al. 2022b; Alshboul et al. 2022a); neural language processing (NLP) (Choi et al., 2022); genetic algorithms (Leigh et al., 2022), random forest (Tuarob et al. 2021; Alshboul et al., 2022a, 2022b; van Steenberg and Mess, 2020; Choi et al. 2021); demand forest, deep learning algorithms (Bodendorf et al. 2022c); data mining (Bodendorf et al. 2022a) and text mining (Biswas et al. 2022) algorithms are widely used in connection with DSS development. Decision trees (Gradient Boosting, Extreme gradient boosting, Adaptive Boosting) is another methodology that we have frequently encountered (Choi et al. 2021; Tuarob et al. 2021; Alshboul et al. 2022b; Shehadeh et al. 2021; Kuo et al. 2021). Multi-agent theory, multi-agent systems and agent-based algorithms (Canelas et al. 2013; Bodendorf et al. 2022c; Kontogounis et al. 2006), and Monte Carlo simulation (Yi et al. 2019) are also witnessed. Undoubtedly, these methodologies and their hybrids can vary according to the requirements of the DSS to be developed. Thus, it is supported that statistical methods—such as regression analysis, hypothesis testing, time series analysis, and Bayesian inference—serve as both precursors and companions to AI/ML approaches, especially being critical in the preprocessing stage of AI/ML pipelines. Outlier detection, correlation analysis, and distribution fitting are generally employed to ensure data quality and relevance. Classical statistical methods also provide baseline models (e.g., linear regression) against which the performance of more complex ML models can be benchmarked. Thus, hybrid models combining statistical and ML approaches are also increasingly applied in forecasting future procurement needs.

Table 4 also supports that along with multi-criteria methods and various statistical approaches, the use of supervised as well as unsupervised algorithms are observed. In the literature, classification of AI and ML methods into supervised methods and unsupervised methods is generally accepted. Commonly mentioned supervised learning algorithms in

Table 4 Distribution according to context, main focus and methodology used

Cluster	Classification	Study	Context	Main focus	Methodology used
Cluster 1 – Review	Overview	Allal-Cherif et al., 2021	Intelligent purchasing with AI	How AI platforms redefine purchasing?	Exploratory study supported by multiple case studies with 5 different technologic platforms (Synertrade, Sillex, SAP Ariba, Jaggaer, Ideapoke)
		Burger et al. 2023	Hybrid intelligence in procurement	AI- Human intelligence interface (advantages, risks, and their effect on procurement performance)	Case studies for order management, demand planning, supplier selection, and procurement analytics using AI elements for different industries, countries and sizes
		Guida et al. 2023	AI's role in procurement	AI in procurement: functions, benefits and challenges	A comprehensive, state-of-the art review.
		Bodendorf et al. 2022b	Supply chain management	Cost-based DS for purchasing in manufacturing companies: ML applications, challenges and obstacles	A Comprehensive review and expert interviews.
Cluster 2 - SCM Processes	Inbound	Reyes-Moro et al. 2003	E-sourcing tools	Development of an e-application for sourcing	AI-supported decision support system development, Fuzzy matching and scoring functions, MAUT (Multi-attribute utility theory), Case study
		Zhou and Chen 2023	Supplier selection	Group decision making with hybrid approach for green supplier selection	Comparison of multicriteria approaches.PF-TOPSIS, PF-VIKOR-MRM, PF-AHP-TOPSIS- LAM combinations PF: Pythagorean fuzzy
		Canelas et al. 2013	Supplier selection	DSS development for supplier selection process to ensure Sarbane-Oxley act compliance	Dynamic learning method and fuzzy knowledge, argumentative negotiation, multi-agent system
		Guillaume et al. 2014	Supplier selection	Supplier risk management with ill-known criteria	Sequential decision making problem under demand and performance indicator uncertainty, decision tree, possibility theory, Hurwies criteria
		Canelas et al. 2015	Supplier contracting process	Development of DSS	Multi-agent approach and dynamic fuzzy learning used for developing a DSS
		Rodrigues-Aquilar et al., 2004	Industrial procurement	DSS for negotiation and analysis of offers	An agent-aware DS service with multiple items and constraints
		Siciliani et al. 2023	Public procurement	Web application development for decision support	NLP, web application development with dashboards

Table 4 (continued)

Cluster	Classification	Study	Context	Main focus	Methodology used
Outbound		Kuo et al. 2021	Smart support system for material procurement	Obtaining predictive big data analytics for procurement	Predictive analytics based on artificial neural networks and decision tree models
		Nissen et al. 2006	Procurement in supply chain	Multi-agent DS development for procurement	Multivariate (MANCOVA) and Univariate (ANCOVA) analysis of variance
		Singh and Tucker 2017	Online product review in e-commerce	Evaluating online customer product reviews via ML to provide purchasing decision support	Multiple ML algorithms comparison and case study
	Overall	Kumar et al. 2022	Online product review in e-commerce	Detection of fraudulent product reviews	Model development based on M-SMOTE algorithm
		Biswas et al. 2022	Online consumer review in e-commerce	On-line consumer reviews	NLP and text mining, text analytics, correlated topic models
		Xu et al. 2011	Consumer electronics	Consumer opinion comparisons from the web	Support vector machine SVM-based method for mining and extracting comparison relations from on-line forums, wikis and blogs
		Bodendorf et al. 2022c	Supply chain management	Collaborative cost management - Buyer supplier dyads	Statistical learning, deep learning, multi-agent theory
Cluster 3 - Sectoral Applications	Manufacturing	Kontogounis et al. 2006	Supply chain management	Development of DSS for multiple SCM processes	Agent-based modular design for SCM processes
		Kazancıoğlu et al., 2023	Supply chain management	SC sustainability and resilience	Fuzzy DEMATEL, Matching AI methods to SC processes via Delphi
		Bodendorf et al. 2022a	Automotive supply chain	ANN for cost estimation	Artificial neural networks, data mining, statistical analysis
		Buyvol et al. 2023	Automobile industry	Diagnostic DSS for defect analysis in automobile industry	Association rules
		Zhang et al. 2021	Electronic toy car production	Big data analysis on sales data using customer purchasing data	Decision-support model development based on clustering big sales data to analyze customer preference analysis. Case study on electric toy cars
		Papachristou et al. 2021	Clothing manufacture	Grouping and combining similar products	Multiple clustering and classification algorithms

Table 4 (continued)

Cluster	Classification	Study	Context	Main focus	Methodology used
Construction		Anglou et al. 2021	Shipping industry	A forecasting tool development to predict expected fleet spare part needs of the fleet for spare parts management	k-means clustering and ML algorithms (random forest, generalized linear model, principal component regression)
		van Steenbergen and Mes 2020	Prelaunch demand forecast	Forecasting new product demand profiles by machine learning	Demand forest, combining quantile regression forest, k-means, and random forest, validated by cases
		Shehadah et al. 2021	Heavy construction equipment	Residual value prediction for heavy construction equipment	Modified decision tree, light GBM, XGBoost regression
		Alshboul et al. 2022a	Heavy construction equipment	DSS development for predicting residual values in heavy construction equipment	Decision support model development based on modelling with deep neural network (DNN), random forest (RF), deep and machine learning regression network integrated with data mining and linear regression (LR) and decision tree (DT)
		Kaklauskas et al. 2007	Construction Industry	On-line system for evaluation of the construction alternatives	A new multi-criteria approach based on COPRAS (method of multi-criteria complex proportional assessment) integrated into a decision support system.
		Kim et al. 2014	Construction Industry	DSS development for optimal material selection	SVM (Support vector machine)
		Kim et al. 2019	Engineering procurement and construction performance	Develops a schedule and cost performance model- DECRIIS (Detailed engineering completion rating index system)	Fuzzy inference system combined with AHP
		Yi et al. 2019	Engineering procurement and construction	Assessment of activities effective on schedule management	Monte Carlo simulation and PM software
		Choi et al. 2021	Engineering procurement and construction	Modular, big-data driven and AI-based support system including multiple processes (bidding, engineering, construction, operations and maintenance)	Variety of algorithms used in combination for different modules (NLP, decision tree, random forest, gradient boosting)
		Choi et al., 2022	Engineering procurement and construction	Contractor risk analysis	Semantic analysis and risk level analysis by NLP and bi-directional short-term memory

Table 4 (continued)

Cluster	Classification	Study	Context	Main focus	Methodology used
Services (Finance, Medical, Air Force)		Alshboul et al. 2022b	Green building construction	Predicting green building costs	Extreme gradient boosting, Deep neural networks, Random forest
		Ahmed and El-Sayegh 2021	Construction Industry	Project delivery methods and criteria in construction sector	Critical and comprehensive review, PRISMA checklist
		Leigh et al. 2002	Stockmarket purchasing	Forecasting prices	Neural network and genetic algorithms for price prediction in purchasing shares.
		Tuarob et al. 2021	Stock market prediction	Real-time stock market prediction: DS tool development for data collection, analysis, and visualization	Extreme gradient boosting, Linear regression, decision tree, k-Nearest neighbours regression, gradient boosting, Bayesian ridge regression, random forest, adaptive boosting.
		Gomes et al. 2019	Medical	Expert system development for early diagnosis of diabetes	Bayesian network, multi-criteria, MAUT (multi-attribute utility theory), MACHBETH, Expert systems development
		Faric et al. 2023	Medical	DSS in radiology	AI-based clinical decision-support considering tool procurement costs
		Owasirikul et al. 2013	Medical	DSS support for aneurysms treatment	Hybrid DSS development combining the machine learning technique with a linear regression
		Caballero et al. 2023	Air force training	Development of a decision support system for pilot training	MMF-DSS (multi-model functional based DSS) functional principal component analysis

Table 5 Distribution by functionalities/processes

Functionalities/Processes	Supply market assessment	Demand prediction and planning	Sourcing planning and forecasting	Supplier searching/scouting	Automated evaluation of tenders/RfQ optimisation	Negotiation support	Supplier selection	Contract management	Order management	Real time incident alerts
Allal-Cherif et al. 2021					X		X			
Burger et al. 2023		X				X	X		X	X
Guida et al. 2023	X		X	X	X	X	X	X		
Bodendorf et al. 2022b										
Reyes-Moro et al. 2003					X	X				
Zhou and Chen 2023										
Canelas et al. 2013					X		X			
Guillaume et al. 2014										
Canelas et al. 2015							X	X		
Rodrigues-Aquilar et al. 2004					X	X				
Siciliani et al. 2023					X			X		
Kuo et al. 2021		X			X			X		
Nissen et al. 2006										
Singh and Tucker 2017										
Kumar et al. 2022										
Biswas et al. 2022										
Xu et al. 2011										
Bodendorf et al. 2022c										
Kontogounis et al. 2006										
Kazancıoğlu et al. 2023		X	X				X			
Bodendorf et al. 2022a					X					
Buyvol et al. 2023										
Zhang et al. 2021										X
Papachristou et al. 2021										
Angiou et al. 2021										
van Steenberghe and Mes 2020		X								
Shehadeh et al. 2021										
Alshboul et al. 2022a										
Kaklauskas et al. 2007							X			
Kim et al. 2014										

Table 5 (continued)

Functionalities/Processes	Supply market assessment	Demand prediction and planning	Sourcing planning and forecasting	Supplier searching/scouting	Automated evaluation of tenders/ RFQ optimisation	Negotiation support	Supplier selection	Contract management	Order management	Real time incident alerts	
Kim et al. 2019											
Yi et al. 2019											
Choi et al. 2022											
Choi et al. 2021											
Alshboul et al. 2022b											
Ahmed and El-Sayegh 2021											
Leigh et al. 2002											
Tuorob et al. 2021											
Gomes et al. 2019											
Faric et al. 2023											
Owasirikul et al. 2013											
Caballero et al. 2023											
Functionalities/Processes	Fraud detection	Procurement analytics	Predictive buying	Purchasing category management	Spend classification and analysis	Cost prediction, modeling and analysis	Inventory management	Maintenance management	Equipment residual life-time prediction	SRM (supplier relationship management)	Supplier performance monitoring and evaluation
Allal-Cherif et al. 2021			X		X					X	X
Burger et al. 2023		X				X					
Guida et al. 2023	X			X	X	X					
Bodendorf et al. 2022b						X					
Reyes-Moro et al. 2003											
Zhou and Chen 2023											
Canelas et al. 2013											
Guillaume et al. 2014											
Canelas et al. 2015											
Rodrigues-Aquilar et al. 2004											
Siciliani et al. 2023											X
Kuo et al. 2021		X									
Nissen et al. 2006											X
Singh and Tucker 2017											

Table 5 (continued)

Functionalities/Processes	Fraud detection	Procurement analytics	Predictive buying	Purchasing category management	Spend classification and analysis	Cost prediction, modeling and analysis	Inventory management	Maintenance management	Equipment residual life-time prediction	SRM (supplier relationship management)	Supplier performance monitoring and evaluation
Kumar et al. 2022											
Biswas et al. 2022											
Xu et al. 2011						X					
Bodendorf et al. 2022c							X				
Kontogounis et al. 2006											
Kazancoglu et al. 2023											
Bodendorf et al. 2022a						X					
Buyvol et al. 2023											
Zhang et al. 2021											
Papachristiou et al. 2021											
Anglou et al. 2021								X			
van Steenberghe and Mes 2020											
Shehadeh et al. 2021									X		
Alshboul et al. 2022a									X		
Kaklauskas et al. 2007									X		
Kim et al. 2014											
Kim et al. 2019											
Yi et al. 2019											
Choi et al. 2022						X					
Choi et al. 2021								X			
Alshboul et al. 2022b						X					
Ahmed and El-Sayegh 2021											
Leigh et al. 2002											
Tuarcob et al. 2021											
Gomes et al. 2019											
Faric et al. 2023											
Owasirikul et al. 2013											
Caballero et al. 2023			X								

Table 5 (continued)

Functionalities/Processes	Supplier risk management	Supply chain finance	Project management support	Procurement sustainability support	Strategic management support	Designer support	Evaluation of customer feedback, reviews and opinions	Big data analysis for sales	Quality failure detection and failure analytics	Material selection	Real time stock market prediction	Early diagnosis and treatment of diseases
Allal-Cherif et al. 2021			X		X							
Burger et al. 2023					X							
Guida et al. 2023		X			X							
Bodendorf et al. 2022b					X							
Reyes-Moro et al. 2003												
Zhou and Chen 2023				X								
Canelas et al. 2013	X											
Guillaume et al. 2014	X				X							
Canelas et al. 2015												
Rodrigues-Aquilar et al. 2004												
Siciliani et al. 2023												
Kuo et al. 2021												
Nissen et al. 2006						X						
Singh and Tucker 2017						X	X					
Kumar et al. 2022						X	X					
Biswas et al. 2022						X	X					
Xu et al. 2011							X					
Bodendorf et al. 2022c												
Kontogounis et al. 2006												
Kazancıoğlu et al. 2023												
Bodendorf et al. 2022a	X			X	X							
Buyvol et al. 2023												
Zhang et al. 2021							X		X			
Papachristou et al. 2021							X					
Anglou et al. 2021							X	X				
van Steenberghe and Mes 2020												
Shehadeh et al. 2021												
Alabboul et al. 2022a												
Kaklauskas et al. 2007										X		
Kim et al. 2014										X		

Table 5 (continued)

Functionalities/Processes	Supplier risk management	Supply chain finance	Project management support	Procurement sustainability support	Strategic management support	Designer support	Evaluation of customer feedback, reviews and opinions	Big data analysis for sales	Quality failure detection and failure analytics	Material selection	Real time stock market prediction	Early diagnosis and treatment of diseases
Kim et al. 2019			X	X								
Yi et al. 2019			X	X								
Choi et al. 2022	X		X	X								
Choi et al. 2021	X		X	X								
Alshboub et al. 2022b				X								
Ahmed and El-Sayegh 2021			X	X								
Leigh et al. 2002											X	
Tuorob et al. 2021											X	
Gomes et al. 2019												X
Faric et al. 2023												X
Owasirikul et al. 2013												X
Caballero et al. 2023												

the literature appear as: k-nearest neighbors, linear regression, logistic regression, decision trees, random forests, support vector machine, gradient boosting and neural networks. Main unsupervised algorithms can be mentioned as k-means clustering, hierarchical clustering, association rule learning and principal component analysis (Wang, 2024). According to this classification, Table 4 reveals that the use of supervised algorithms is more dominant, with various forms of regression, neural networks and decision trees. Unsupervised algorithms are also seen in the table, such as k-means clustering, association rule learning and principal component analysis.

Although the difference between supervised and unsupervised learning is clear, this distinction can be blurred with the increasing use of contemporary hybrid approaches, semi-supervised, self-supervised and deep learning algorithms (Pröve 2018; Brownlee 2019). Such algorithms blur the line since they are not purely supervised or unsupervised (Garg and Kalai, 2017). This makes it difficult to make clear-cut categorizations, which is especially true for classification and clustering. Thus, the authors avoided including such a classification in the tables.

7 Analysis by functionalities/processes

Our study showed that a multitude of processes and subprocesses are supported by the AI-ML-DS combination. Table 5 shows the distribution by functionalities/processes.

The table indicates that along with demand prediction and planning (Burger et al. 2023), various sub-steps of procurement (supply market assessment, sourcing, demand planning and forecasting, supplier searching/scouting, negotiation support, supplier selection, tender/RFQ process, contract management) are found to be frequently supported by AI, ML and DS (Guida et al. 2023; Kontogounis et al. 2006). Predictive buying and real-time procurement analytics are enabled for the end-to-end procurement cycle with overall order management cycle automated (Caballero et al. 2023). Decision support and advanced reporting for purchasing category management, spend classification and analysis analytics are facilitated, appearing as one of the greatest contributions of AI and ML to procurement.

It can also be argued that sustainability of the end-to-end procurement process is well-assisted by AI and ML in connection with DSS systems. Automated evaluation of tenders/RFQ optimization, supplier selection, cost prediction modelling and analysis appear as significant sub-processes benefiting from these technologies (Kontogounis et al., 2006).

We have also witnessed the use of ML and AI in combination with DSS in maintenance management (Anglou et al. 2021), inventory management (Kontogounis et al., 2006), supplier performance monitoring, SRM, supply chain risk management (Choi et al. 2021; Choi et al., 2022), and quality management. Predictive capabilities are increased, incident reporting and alerting is enabled, and customer reviews and feedback are better analyzed. Overall project management cycle (Kim et al. 2019; Yi et al. 2019; Choi et al. 2021; Choi et al., 2022) and sustainability of the overall procurement cycle are also well-supported.

Consequently, it is disclosed that besides significantly supporting the procurement process, AI/ML usage in connection with the DSS is beneficial for many other SC processes and assist in end-to-end project management cycle.

8 Benefits and challenges of using AI and ML for procurement&purchasing decision support

8.1 Benefits

This section overviews the benefits of AI and ML utilization for procurement&purchasing decision-support from both academic and business case perspectives. Identified benefits with their corresponding academic references are given in Table 6.

Our findings reveal that major benefits are automation and optimization of procure-to-pay cycle (Allal-Cherif et al., 2021; Burger et al. 2023; Guida et al. 2023; Canelas et al. 2015; Siciliani et al. 2023); ensuring synchronous communication and information (Bodendorf et al. 2022b; Reyes-Moro et al. 2003); improved monitoring and control as well as visibility and transparency (Allal-Cherif et al., 2021; Burger et al. 2023; Guida et al. 2023; Bodendorf et al. 2022b; Reyes-Moro et al. 2003). Concrete real-life examples can be mentioned as Basware's AI-driven invoice automation reducing manual interventions (Basware 2024), Unilever's AI-based spend analytics improving the transparency in terms of spend categories (AI Expert Network 2023a), and GE Aviation's application to detect purchasing anomalies and unusual purchasing behaviour (DigitalDefynd 2025).

Minimized delays at various points of the procurement cycle (Allal-Cherif et al., 2021), reduced cycle time (Guida et al. 2023; Reyes-Moro et al. 2003; Papachristou et al. 2021), improved accuracy in planning, control and execution (Allal-Cherif et al., 2021; Guida et al. 2023; Kim et al. 2019; Yi et al. 2019) and more efficient procurement decisions (Allal-Cherif et al., 2021) are well-supported benefits identified from the set of studies involved. An example can be given as Siemens' automated demand forecasting to anticipate the inventory needs and reduce the procurement delays (AI Expert Network 2023b). An overall procure-to-pay compliance is also ensured (Guida et al. 2023; Canelas et al. 2013, 2015). SAP Ariba's AI-driven compliance engine is a major example for real-time detection of contract deviations and regulatory risks (Accenture 2024).

All these identified benefits have direct positive effects on the overall SCM. Better partnership creation with critical suppliers with better alignment of internal and external partner systems (Guida et al. 2023), and improved supply chain relations and collaboration (Guida et al. 2023; Guillaume et al. 2014; Bodendorf et al. 2022b; Papachristou et al. 2021) became possible. Towards supplier relations, Procter&Gambler utilizes AI for monitoring strategic supplier KPI's for long-term supplier collaboration (AI Expert Network 2023c), and Dell utilizes ML to enhance supplier collaboration portals to enable seamless data transfer (Davis 2010). Johnson & Johnson's AI-monitored supplier defect rate metrics (Telang 2024) and Amazon's AI-supported real-time supply plan and order changes (Krause 2024) are further examples. Such applications lead to better adaptation capabilities, improved supplier and procurement performance, and risk reduction from SCM perspective (Guida et al. 2023; Guillaume et al. 2014; Bodendorf et al. 2022a; Kim et al. 2019; Yi et al. 2019).

Therefore, it appears that AI and ML utilization brings huge benefits to the procurement process, the core process governing the SC relations. Both academic references and business case examples implemented by the giant enterprises support these benefits. This clearly demonstrates the applicability of AI-ML assistance from a practical perspective. Improvements obtained in the radically transformed and automated procurement processes result in improved and healthier supply-side relations and partnerships with suppliers. Undoubtedly,

Table 6 Benefits of using AI and ML for procurement&purchasing decision support

Benefits	Automation and optimisation of procure-to-pay cycle	Synchronous communication and information	Improved monitoring and control	Improved visibility and transparency	Minimized delays	Reduced cycle time	Improved accuracy in planning, control and execution	More efficient procurement decisions	Procure-to-pay compliance
Allal-Cherif et al. 2021	X	X	X	X	X		X	X	
Burger et al. 2023	X	X	X	X					
Guida et al. 2023	X	X	X	X		X	X		X
Bodendorf et al. 2022b		X	X	X					
Reyes-Moro et al. 2003		X	X	X		X			
Canelas et al. 2013			X	X					X
Guillaume et al. 2014									
Canelas et al. 2015	X								X
Siciliani et al. 2023	X								
Singh and Tucker 2017	X								
Kumar et al. 2022	X								
Biswas et al. 2022	X								
Bodendorf et al. 2022c									
Kontogounis et al. 2006									
Kazancoglu et al. 2023				X					
Bodendorf et al. 2022a				X					
Buyvol et al. 2023	X								
Papachristou et al. 2021						X			
Kim et al. 2014									
Kim et al. 2019							X		
Yi et al. 2019							X		
Choi et al. 2022			X						
Choi et al. 2021			X						

Table 6 (continued)

Benefits	Partnership creation with critical suppliers	Better alignment of internal systems and external partners systems	Improved supply chain relations/ collaboration	Real time adaptation to external requests	Improved supplier performance	Improved procurement performance	Risk reduction in supply chain	Customized design	Providing hybrid intelligence
Allal-Cherif et al. 2021									
Burger et al. 2023									X
Guida et al. 2023	X	X	X	X	X	X	X		
Bodendorf et al. 2022b									
Reyes-Moro et al. 2003									
Canelas et al. 2013									
Guillaume et al. 2014			X				X		
Canelas et al. 2015									
Siciliani et al. 2023									
Singh and Tucker 2017									
Kumar et al. 2022									
Biswas et al. 2022									
Bodendorf et al. 2022c			X						
Kontogounis et al. 2006					X	X			
Kazancoglu et al. 2023									
Bodendorf et al. 2022a							X		
Buyvol et al. 2023									
Papachristou et al. 2021			X					X	
Kim et al. 2014								X	
Kim et al. 2019									
Yi et al. 2019									
Choi et al. 2022							X		
Choi et al. 2021							X		

all these benefits serve for improved SC collaboration, performance, responsiveness, and adaptiveness of the overall network.

Consequently, the main practical insight for procurement managers is that any AI/ML investment in AI/ML go far beyond the operational efficiencies. AI capabilities should be utilized for all the above-mentioned benefits regarding automation, visibility, control, cycle time reductions, end-to-end management as well as boosting supplier relations and collaboration.

8.2 Challenges

Integrating ML and AI support into the existing enterprise application infrastructure comes with numerous challenges which require handling technical, organizational and managerial aspects.

Creating knowledge and awareness of technology is among the most critical and commonly cited challenges (Guida et al. 2023; Bodendorf et al. 2022b). People at all positions should be aware of the potential of these new disruptive technologies and receive necessary training (Fries et al. 2025). In this regard, understanding the workforce skill requirements and the skill gaps, providing hands-on experience, and ensuring the availability and continuity of the internal analytical skills are vital. Creating a continuous learning environment and fostering collaborative learning culture with on-going mentorship are crucial, including supplier training and engagement (Van Hoek 2024; Fries et al. 2025).

While fostering awareness, skills and continuous learning environment, practical implementations are often hindered by a set of complex technological challenges. In this regard, major technological challenges involve the following:

- (a) Lack of systemic data availability due to existing fragmented or siloed IT infrastructures, problems with data accuracy and completeness, data quality and overall data management processes are critical issues (Jawad and Balazs 2024; Bodendorf et al. 2022b; Mhaskey 2024). They complicate the extract-transform-load processes, hinder the training of the AI algorithms, impair the predictions and affect the overall reporting and decision-support process (Narne 2022). Ensuring interoperability and real-time continuous data processing is essential for dynamic and accurate results (Javidrooz et al. 2020).
- (b) Integration with the existing information system components (legacy systems, backbone ERP systems containing the core processes; SRM systems or other IoT components) is required (Guida et al. 2023; Van Hoek 2024; Narne 2022). Compatibility of the data infrastructures of the partners is essential, and integrity of the add-on AI and ML modules with the existing partner information systems must be ensured by handling and incompatibilities and protocol mismatches. In this regard, utilizing middleware and API's (Application Programming Interface) of various forms appear as the main technological solutions. Immature IS foundation will definitely hinder the integration of any further AI/ML application.
- (c) Systems performance issues of algorithmic complexity, latency, throughput and scalability affect the current performance as well as future growth and expandability of the system (Mhaskey 2024).

- (d) Data security and privacy are among major concerns. Utilization of appropriate data encryption approaches, applying comprehensive data governance strategies, ensuring clear access control policies with appropriate identity management and on-going monitoring for data quality and compliance appear as the main solutions to handle this set of challenges.
- (e) Ensuring the acceptance of the AI or ML model to be integrated into the system and assuring confidence in the model results are major issues. Since the usage of AI and ML support requires various modelling approaches integrated into the enterprise backbone and decision-support system, validation and verification of the related models appear vital for ensuring confidence in the results obtained (Bodendorf et al. 2022b).

Similar to other management information system projects and ERP implementations, business process reengineering is among the top issues, acting as a prerequisite to enterprise systems integration (Laudon and Laudon 2022; Javidroozi et al. 2020). The absence of existing well-structured procurement processes is the most critical drawback, requiring business process reengineering (BPR) of both within-enterprise and cross-partner processes with detailed process analysis and mapping for obtaining streamlined and seamlessly integrated processes.

Applying a systematic project management discipline is inevitable for success, and to keep the scope, schedule, cost and quality of the project under control represents a significant challenge (Laudon and Laudon 2022). Managing time and cost of implementation-related constraints with realistic schedules while measuring progress and obtaining economic efficiency are among the serious concerns (Lam 2005; Bodendorf et al. 2022b; Guida et al. 2023). Initial investment concerns and all project-related cost items, including implementation, should be handled within a project management perspective. In this regard, ensuring cost transparency is vital across the entire project (Bodendorf et al. 2022b).

Along with the technical issues, handling the managerial and organizational aspects is vital. Offering proper leadership, ensuring commitment, support, approval and prioritization of top management and employee engagement are essential, which offer separate set of challenges (Laudon and Laudon 2022; Jones 2012) and well-proven in the literature from ERP implementations, organizational change and management information systems. The importance of handling change management issues and coping with resistance to change are also the issues that are never to be neglected (Lam 2005; Mhaskey 2024; Guida et al. 2023; Laudon and Laudon 2022).

The absence of AI governance frameworks and the need for handling regulatory and compliance issues with proper documentation further represent a separate set of challenges. Ensuring policies about accountability, explainability, trust, auditability and transparency, as well as ethical use and privacy are essential (Narne 2022; Mhaskey 2024).

It is important to highlight that challenges identified are totally in line with the management information systems and ERP implementation literature for managing the enterprise-wide IT-based transformation projects. Key managerial insight is that integrating AI and ML solutions to the enterprise application systems architecture requires systematic BPR and project management efforts while attending all the hard and soft aspects.

9 Discussion and further research avenues

Our study supported that disruptive technologies of AI and ML are particularly suited to procurement, which is the key process for SC collaboration and partnership. AI and ML facilitate procurement decision-support for almost all subprocesses. Various stages of procurement, including strategic sourcing, RfQ collection and evaluation, supplier selection and order management are assisted by the AI and ML tools and algorithms. End-to-end automation, advanced decision-support and built-in self-training and self-learning capabilities are embedded at different stages of the procurement cycle with a diversity of AI and ML applications.

The study revealed the co-existence of statistical and MCDM methods with AI and ML methods, both independently and jointly. Co-occurrence and the hybrid use of AI-ML and MCDM approaches are shown to combine the strengths of both sides to provide dynamism and automated procurement&decision-support in data-intensive environments. The synergy between AI/ML algorithms and statistical methods establishes a robust analytical foundation for procurement& purchasing decision-support systems. Statistical techniques ensure rigor, interpretability, and foundational modeling while AI and ML enable increased adaptivity, enhanced predictive capabilities and scalability. Jointly, they enable data-driven procurement strategies that are more responsive, intelligent, and aligned with organizational goals in increasingly complex supply environments. Thus, integrating a wide spectrum of AI and ML tools and algorithms into DSS designs in close integration with MCDM and statistical approaches will provide us with unprecedented levels of forecasting/estimation, prediction, controlling, and decision-support capabilities at all subprocesses and decision levels, enabling fact-based and accelerated decision-making. Hence, the real issue is ensuring proper integration of AI and ML with the DSS designs and ERP backbones of the enterprises. With successful integration, such systems will become a boosting factor for the overall SCM and elevate collaborative decision-making across the supply network to entirely different levels.

The study revealed that in recent literature benefits are studied in much more detail when compared with the challenges involved. A variety of applications in the construction sector, involving significant spectrum of procurement activities, also became apparent.

It is worth highlighting that we have spotted the study in Guida et al. 2023 as one of the most comprehensive and recent studies covering multiple aspects of the topic. Guida et al. 2023 mention that their review covers studies until December 2020. Our study included no date range filtering and covered more recent studies corresponding to the years containing the highest number of studies regarding our keyword intersection (as supported by the graph in Fig. 2). With all the comprehensive summary analytics, taxonomy tables and graphs, our review provided a recent and broad coverage.

Our study reveals many further research avenues, which can be summarized as below:

- a) **Implementation Challenges:** As supported by Guida et al. 2023; implementation challenges should be studied further. Case-based studies in relation to technological, managerial, and organizational challenges involved will be valuable contributions. Further investigation of success and failure factors along with best practices is also required via case-based studies. Future research can explore the research questions such as: “how technological challenges can be handled for AI and ML integration?”; and “what are the main success factors in AI and ML utilization for procurement process?”

- b) ERP Integration: Connectivity and integrity of AI and ML tools with the existing ERP backbones appear vital. Conceptual modelling efforts as well as applications will be a significant contribution in this direction. Developing different generic configurations using AI and ML in hybrids using other IoT technologies are needed. Representative research questions can be: “how can IoT technologies be integrated with AI and ML tools?”; and “how can blockchain technology be integrated with AI and ML in procurement?”
- c) Spend Analysis and Reporting: Spend management with effective analytics is a vital issue in handling the procurement side. Our study clearly revealed the research gap in this direction. Further studies integrating AI and ML with spend analytics and reporting are required at both enterprise and SC levels. Integration of these advanced approaches within a business intelligence perspective represents a still open research avenue. Some suggested research questions can be “how can AI and ML methods contribute to spend analytics and reporting?”, and “what are the successful implementation cases of AI and ML in spend analytics?”
- d) SC Fraud Detection and Integration with Blockchain: Utilization of AI-ML for fraud detection in any SCM process (procurement or other) also holds significant further research potential. This line of research is especially promising if considered together with the use of blockchain technology to prevent fraudulent acts among SC partners. Blockchain technology provides immutable records on a secure and trusted transaction infrastructure. Its potential in the prevention of fraud and counterfeiting across SCM partners is well-proven. In this regard, we reckon that the intersection of blockchain and AI/ML topics hold significant further research potential. Some guiding research questions can be: “what is the role and importance of blockchain in procurement fraud detection?” “how can smart contracts contribute to fraud detection?”
- e) SC Risk management: Further research combining AI-ML with SC risk management will support collaborative and proactive risk management processes with advanced predictive capabilities across partners. Some proposed research can be “how can AI and ML methods be used for mitigating procurement risks?” and “How can AI-ML methods contribute to real-time risk monitoring and follow-up?”

10 Conclusion

This study put forward a comprehensive literature review at the intersection of AI-ML-procurement. The analysis reveals various existing studies focusing on general overview of the domain, focusing on either ML or AI, investigating different sectoral applications in specific contexts or scenarios, dealing with only specific individual subprocesses/subfunctions of the procurement process (such as supplier searching, negotiation, contract management), or applying specific methodologies.

This review is differentiated from the other studies, and thus a unique contribution to the literature due to the following:

- providing simultaneous treatment of AI, ML and procurement concepts instead of treating them individually,
- having a broad process-centric approach to procurement, covering its multiple subproc-

esses and functionalities,

- offering detailed taxonomic analysis in multiple dimensions (approaches and methodologies used, SCM processes/functionalities covered, sectoral breakdown of the applications, and benefits),
- suggesting multiple further research avenues.

It became evident that AI and ML already started to revolutionize procurement process and related decision-support mechanisms. Benefits identified from the academic studies and the real-life case examples from the giant enterprises supported the overall potential and assimilation from the procurement perspective. As supported by the other studies, we reckon that AI and ML diffusion will continue in procurement and purchasing domain with an increasing pace. In this regard, guaranteeing a balance of AI/ML and human intelligence to find the correct hybrids at different SCM processes will boost the overall performance. The above section has already put forward multiple promising research avenues. It also became evident that hybrid use of AI-ML with other Industry 4.0 technologies is not mature.

Along with the open research avenues offered, benefits and challenges identified in this study are useful for further applications to practitioners and procurement people, providing guidance to especially procurement managers. We also believe that our study sheds light to the idea of integrating advanced AI and ML tools and algorithms into various existing ERP and DSS infrastructures of the enterprises.

Our study is limited with the search keywords and filtering mechanisms used. Although basing the search and filtering on the WoS database appears as a limitation, this enabled us to eliminate the risk of including inconsistent quality resources and to provide a clear and focused coverage. Undoubtedly, the study can be repeated by expanding the scope to include other databases and eliminating the duplicate studies that are already covered by WoS. By the selection of different search keywords in the review systematic, an array of applications in various service sectors can also be investigated. Due to the multiplicity and wide spectrum of existing AI and ML methods, it was not feasible to offer a detailed analysis for specific methods, which may also be considered as a limitation. As mentioned in the further research avenues, studies focusing on implementation challenges, ERP integration, spend analysis and reporting, SC fraud detection and integration with blockchain as well as SC risk management deserve separate further studies.

Author contributions Both authors have equal contribution on conceptualization, methodology, formal analysis, writing (original draft preparation, review and editing), supervision and project administration.

Funding The authors declare that no funds, grants, or other support was received.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it.

The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

- Accenture (2024) Journey to intelligent procurement with SAP Ariba. Retrieved April 29, 2025, from <https://www.accenture.com/ro-en/case-studies/about/journey-intelligent-procurement-sap-ariba>
- Ahmed S, El-Sayegh S (2021) Critical review of the evolution of project delivery methods in the construction industry. *Buildings* 11(1):11. <https://doi.org/10.3390/buildings11010011>
- AI Expert Network (2023)a. Case study: Unilever's integration of AI in the supply chain. 17 August 2023, Retrieved April 29, 2025, from <https://aiexpert.network/case-study-unilevers-integration-of-ai-in-the-supply-chain/>
- AI Expert Network (2023)b. Case study: How Siemens is transforming supply chain with AI. 15 August 2023 Retrieved April 29, 2025, from <https://aiexpert.network/case-study-how-siemens-is-transforming-supply-chain-with-ai/>
- AI Expert Network (2023)c. Case study: How AI is transforming Procter & Gamble's global operations. 31 July 2023, Retrieved April 29, 2025, from <https://aiexpert.network/case-study-how-ai-is-transforming-procter-gambles-global-operations/>
- Allal-Chérif O, Simón-Moya V, Ballester ACC (2021) Intelligent purchasing: how artificial intelligence can redefine the purchasing function. *J Bus Res* 124:69–76. <https://doi.org/10.1016/j.jbusres.2020.11.050>
- Alshboul O, Shehadeh A, Al-Kasasbeh M, Mamlook A, Halalsheh RE, Alkasasbeh N, M (2022a) Deep and machine learning approaches for forecasting the residual value of heavy construction equipment: a management decision support model. *Eng Constr Architectural Manage* 29(10):4153–4176. <https://doi.org/10.1108/ECAM-08-2020-0614>
- Alshboul O, Shehadeh A, Almasabha G, Almuflih AS (2022b) Extreme gradient boosting-based machine learning approach for green Building cost prediction. *Sustainability* 14(11):6651. <https://doi.org/10.3390/su14116651>
- Anglou FZ, Ponis S, Spanos A (2021) A machine learning approach to enable bulk orders of critical spare-parts in the shipping industry. *J Ind Eng Manag* 14(3):604–621. <https://doi.org/10.3926/jiem.3446>
- Basware (2024) AI transforms financial operations at Billerud: Case study. Retrieved April 29, 2025, from <https://customer.basware.com/en/ai-transforms-financial-operations-at-billerud>
- Biswas B, Sengupta P, Kumar A, Delen D, Gupta S (2022) A critical assessment of consumer reviews: a hybrid NLP-based methodology. *Decis Support Syst* 159:113799. <https://doi.org/10.1016/j.dss.2022.113799>
- Bodendorf F, Lutz M, Michelberger S, Franke J (2022a) An empirical investigation into intelligent cost analysis in purchasing. *Supply Chain Management- Int J* 27(6):785–808. <https://doi.org/10.1108/SCM-11-2020-0563>
- Bodendorf F, Merkl P, Franke J (2022b) Artificial neural networks for intelligent cost estimation - a contribution to strategic cost management in the manufacturing supply chain. *Int J Prod Res* 60(21):6637–6658. <https://doi.org/10.1080/00207543.2021.1998697>
- Bodendorf F, Xie Q, Merkl P, Franke J (2022c) A multi-perspective approach to support collaborative cost management in supplier-buyer dyads. *Int J Prod Econ* 245:108380. <https://doi.org/10.1016/j.ijspe.2021.108380>
- Brownlee J (2019) 14 Different types of learning in machine learning. Retrieved April 30 2025, from: <https://machinelearningmastery.com/types-of-learning-in-machine-learning/>
- Burger M, Nitsche AM, Arlinghaus J (2023) Hybrid intelligence in procurement: disillusionment with ai's superiority? *Comput Ind* 150:103946. <https://doi.org/10.1016/j.compind.2023.103946>
- Buyvol P, Makarova I, Voroshilov A, Krivonogova A (2023) The process of identifying automobile joint failures during the operation phase: data analytics based on association rules. *Information* 14(5):257. <https://doi.org/10.3390/info14050257>
- Caballero WN, Gaw N, Jenkins PR, Johnstone C (2023) Toward automated instructor pilots in legacy air force systems: physiology-based flight difficulty classification via machine learning. *Expert Syst Appl* 231:20711. <https://doi.org/10.1016/j.eswa.2023.120711>
- Canelas JAF, Martin QM, Rodriguez JMC (2013) Argumentative SOX compliant and quality decision support intelligent expert system over the suppliers selection process. *Appl Comput Intell Soft Comput* 2013(973704). <https://doi.org/10.1155/2013/973704>

- Canelas JAF, Martin QM, Rodriguez JMC (2015) Argumentative SOX compliant and intelligent decision support systems for the suppliers contracting process. *Intell Techniques Eng Management: Theory Appl* 87:333–375. https://doi.org/10.1007/978-3-319-17906-3_14
- Choi SW, Lee EB (2022) Contractor's risk analysis of engineering procurement and construction (epc) contracts using ontological semantic model and bi-long short-term memory (LSTM) technology. *Sustainability* 14(11):6938. <https://doi.org/10.3390/su14116938>
- Choi SW, Lee EB, Kim JH (2021) The engineering machine-learning automation platform: a big-data-driven AI tool for contractors' sustainable management solutions for plant projects. *Sustainability* 13(18):10384. <https://doi.org/10.3390/su131810384>
- Davis M (2010) Case study for supply chain leaders: Dell's transformative journey through supply chain segmentation. Gartner Research. ID Number G, 208603. Retrieved April 30, 2025, from <https://www.gartner.com/document/208603>
- DigitalDefynd (2025) Top 30 machine learning case studies. Retrieved April 29, 2025, from <https://digitaldefynd.com/IQ/machine-learning-case-studies/>
- Faric N, Hinder S, Williams R, Ramaesh R, Bernabeu MO, van Beek E, Cresswell K (2023) Early experiences of integrating an artificial intelligence-based diagnostic decision support system into radiology settings: a qualitative study. *J Am Med Inform Assoc* 31(1):24–34. <https://doi.org/10.1093/jamia/ocad191>
- Fries M, Nießner J, Ludwig T, Kotthaus C (2025) Exploring AI integration in SME production planning: design spaces and the role of workers. *Comput Supported Coop WorkK (CSCW)* Early Access. <https://doi.org/10.1007/s10606-025-09512-6>. published on-line: 3. April 2025
- Garg VK, Kalai AT (2016) Meta-Unsupervised-Learning: A supervised approach to unsupervised learning. Retrieved April 29, 2025 from: *arXiv preprint arXiv:1612.09030*
- Gomes E, Pinheiro PR, Pinheiro MCD, Nunes LC, Gomes LBG (2019) Heterogeneous methodology to support the early diagnosis of gestational diabetes. *IEEE Access* 7:67190–67199. <https://doi.org/10.1109/ACCESS.2019.2903691>
- Guida M, Caniato F, Moretto A, Ronchi S (2023) The role of Artificial intelligence in the procurement process: state of the Art and research agenda. *J Purchasing Supply Manage* 29(2):100823. <https://doi.org/10.1016/j.pursup.2023.100823>
- Guillaume R, Marques G, Thierry C, Dubois D (2014) Decision support with ill-known criteria in the collaborative supply chain context. *Eng Appl Artif Intell* 36:1–11. <https://doi.org/10.1016/j.engappai.2014.06.013>
- Hendriksen C (2023) Artificial intelligence for supply chain management: disruptive innovation or innovative disruption? *J Supply Chain Manage* 59(3):65–76
- Javidroozi V, Shah H, Feldman G (2020) A framework for addressing the challenges of business process change during enterprise systems integration. *Bus Process Manage J* 6:2
- Jawad ZN, Balazs V (2024) Machine learning-driven optimization of enterprise resource planning (ERP) systems: a comprehensive review. *J Basic Appl Sci* 13(4):1–13
- Jones GR (2012) *Organizational Theory, Design and Change*. Pearson
- Jonsson P (2008) *Logistics and supply chain management*. McGraw-Hill: Berkshire
- Kaklauskas A, Zavadskas EK, Trinkunas V (2007) A multiple criteria decision support on-line system for construction. *Eng Appl Artif Intell* 20(2):163–175. <https://doi.org/10.1016/j.engappai.2006.06.009>
- Kazancoglu I, Ozbiltekin-Pala M, Mangla SK, Kumar A, Kazancoglu Y (2023) Using emerging technologies to improve the sustainability and resilience of supply chains in a fuzzy environment in the context of COVID-19. *Ann Oper Res* 322(1):217–240. <https://doi.org/10.1007/s10479-022-04775-4>
- Kim J, Kim S, Tang L (2014) Case study on the determination of Building materials using a support vector machine. *J Comput Civil Eng* 28(2):315–326. [https://doi.org/10.1061/\(ASCE\)CP.1943-5487.0000259](https://doi.org/10.1061/(ASCE)CP.1943-5487.0000259)
- Kim MH, Lee EB, Choi HS (2019) A forecast and mitigation model of construction performance by assessing detailed engineering maturity at key milestones for offshore Epc mega-projects. *Sustainability* 11(5):1256. <https://doi.org/10.3390/su11051256>
- Kontogounis I, Chatzidimitriou KC, Synaenidis AL, Mitkas PA (2006) A robust agent design for dynamic SCM environments. *Adv Artif Intell* 3955:127–136
- Krause C (2024) Case study: Amazon's AI-driven supply chain: a blueprint for the future of global logistics. *The CDO Times*. Retrieved April 30, 2025 from <https://cdotimes.com/2024/08/23/case-study-amazons-ai-driven-supply-chain-a-blueprint-for-the-future-of-global-logistics/>
- Kumar A, Gopal RD, Shankar R, Tan KH (2022) Fraudulent review detection model focusing on emotional expressions and explicit aspects: investigating the potential of feature engineering. *Decis Support Syst* 155:113728. <https://doi.org/10.1016/j.dss.2021.113728>
- Kuo TC, Peng CY, Kuo CJ (2021) Smart support system of material procurement for waste reduction based on big data and predictive analytics. *Int J Logist Res Appl*. <https://doi.org/10.1080/13675567.2021.1969348>

- Lam W (2005) Investigating success factors in enterprise application integration: a case-driven analysis. *Eur J Inform Syst* 14(2):175–187
- Laudon KC, Laudon JP (2022) *Management information systems: managing the digital firm*. Pearson, Global Edition
- Leigh W, Purvis R, Ragusa JM (2002) Forecasting the NYSE composite index with technical analysis, pattern recognizer, neural network, and genetic algorithm: a case study in decision support. *Decis Support Syst* 32(4):361–377. [https://doi.org/10.1016/S0167-9236\(01\)00121-X](https://doi.org/10.1016/S0167-9236(01)00121-X)
- Mhaskey SV (2024) Integration of artificial intelligence (AI) in enterprise resource planning (ERP) systems: opportunities, challenges, and implications. *Int J Comput Eng Res Trends* 11(12):1–9
- Narne H (2022) AI and machine learning in enterprise resource planning: empowering automation, performance, and insightful analytics. *Int J Res Anal Reviews* 9(1):284–288
- Nissen ME, Sengupta K (2006) Incorporating software agents into supply chains: experimental investigation with a procurement task. *MIS Q* 30(1):145–166
- Owasirikul W, Tantivatana J, Gansawat D, Auethavekiat S (2013) Prediction of shape diameter undergoing coil embolization of saccular intracranial aneurysm treatment using a hybrid decision support system. *Australasian Phys Eng Sci Med* 36(2):177–191. <https://doi.org/10.1007/s13246-013-0193-1>
- Papachristou E, Chrysopoulos A, Bilalis N (2021) Machine learning for clothing manufacture as a mean to respond quicker and better to the demands of clothing brands: a Greek case study. *Int J Adv Manuf Technol* 115(3):691–702. <https://doi.org/10.1007/s00170-020-06157-1>
- Pröve PL (2018) The blurry lines of supervised and unsupervised learning. <https://medium.com/data-science/the-blurry-lines-of-supervised-and-unsupervised-learning-b8a2aa04c8b0>
- Reyes-Moro A, Rodríguez-Aguilar JA, López-Sánchez M, Cerquides J, Gutierrez-Magallanes D (2003) Embedding decision support in e-sourcing tools: quotes, a case study. *Group Decis Negot* 12(4):347–355. <https://doi.org/10.1023/A:1024824005214>
- Richey RG Jr, Chowdhury S, Davis-Sramek B, Giannakis M, Dwivedi YK (2023) Artificial intelligence in logistics and supply chain management: a primer and roadmap for research. *J Bus Logist* 44(4):532–549
- Rodríguez-Aguilar JA, Reyes-Moro A, Giovanucci A, Cerquides J, Noria FX (2004) Negotiation support in highly-constrained trading scenarios. *Curr Top Artif Intell* 3040:425–434
- Shehadeh A, Alshboul O, Mamlook A, Hamedat RE, O (2021) Machine learning models for predicting the residual value of heavy construction equipment: an evaluation of modified decision tree, lightgbm, and XGBoost regression. *Autom Constr* 129:103827. <https://doi.org/10.1016/j.autcon.2021.103827>
- Siciliani L, Taccardi V, Basile P, Di Ciano M, Lops P (2023) AI-based decision support system for public procurement. *Inform Syst*. <https://doi.org/10.1016/j.is.2023.102284>. 119,102284
- Singh A, Tucker CS (2017) A machine learning approach to product review disambiguation based on function, form and behavior classification. *Decis Support Syst* 97:81–91. <https://doi.org/10.1016/j.dss.2017.03.007>
- Telang A (2024) Artificial intelligence at Johnson & Johnson. 21 October 2024, Retrieved April 30, 2025, from <https://emerj.com/artificial-intelligence-at-johnson-and-johnson/>
- Tuarob S, Wettayakorn P, Phetchai P, Traivijitkhun S, Lim S, Noraset T, Thaipisutikul T (2021) DAVIS: a unified solution for data collection, analyzation, and visualization in real-time stock market prediction. *Financial Innov* 7(1):56. <https://doi.org/10.1186/s40854-021-00269-7>
- van Hoek R (2024) Insight from industry-early lessons learned about AI adoption in core procurement processes, directions for managers and researchers. *Supply Chain Management: Int J* 29(4):794–803
- van Steenberg RM, Mes MRK (2020) Forecasting demand profiles of new products. *Decis Support Syst* 139:113401. <https://doi.org/10.1016/j.dss.2020.113401>
- Wang W (2025) *Principles of machine Learning: the three perspectives*. Springer
- Xu KQ, Wang W, Ren JSJ, Xu J, Liu L, Liao SSY (2011) Classifying consumer comparison opinions to uncover product strengths and weaknesses. *Int J Intell Inf Technol* 7(1):1–14. <https://doi.org/10.4018/jiit.2011010101>
- Yi D, Lee EB, Ahn J (2019) Onshore oil and gas design schedule management process through time-impact simulations analyses. *Sustainability* 11(6):1613. <https://doi.org/10.3390/su11061613>
- Zhang J, Chu XP, Simeone A, Gu PH (2021) Machine learning-based design features decision support tool via customers purchasing data analysis. *Concur Eng Res Appl* 29(2):124–141. <https://doi.org/10.1177/1063293X20963313>
- Zhou F, Chen TY (2023) A hybrid group decision-making approach involving pythagorean fuzzy uncertainty for green supplier selection. *Int J Prod Econ* 261:108875. <https://doi.org/10.1016/j.ijpe.2023.108875>